



Umoja Hardware System

How the system works — in diagrams

A multi-branch ERP for a hardware retailer: it follows goods from the supplier into the store and out to the customer, and follows the money both ways. Every screen belongs to a role, and every role sees only its own work.

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business modules

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shared database

1 What the system covers
the four areas of the business

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3 Goods in: purchase to stock
ordering, delivery checks, admin sign-off

4 Goods out: sale to delivery
quote, sale, approval, dispatch

5 Who does what
the roles and their daily work

What the system covers

Four areas of the business, one database. A movement recorded in one area shows up in the others.

Goods in	Goods out	Money	People
Suppliers who we buy from	Quotations prices offered to a customer	Expenses & income with receipts attached	Users & roles who may do what
Purchase orders what we ordered, at what price	Sales (POS) cash, credit, part-deposit	Debtors what customers still owe	Employees records, documents, contracts
Delivery checks what actually arrived	Order approval a manager releases the order	Supplier payments what we owe, what we paid	Attendance & leave who is in, who is off
Goods received notes store confirmation	Dispatch vehicle, driver, delivery note	Taxes VAT, PAYE, SDL and the rest	Payroll NSSF, NHIF, PAYE, net pay
Stock per branch live quantity, low-stock alerts	Customers who we sell to	CRM register customer trade history, credit	Performance reviews and discipline

Underneath all four: one shared spine

Branches

every stock figure, sale and expense belongs to a branch

Roles

the same screen shows different things to different roles

Audit trail

who changed what, and when

Live updates

stock and new sales appear without a refresh

How the pieces are wired

Three ways in, one server, one database. The web pages and the phone app use the same API.

WHO CONNECTS

Office browser

Full system. Every role works here.
Server-rendered pages + JavaScript.

Android app

Field sales: POS, stock look-up, customers, quotations. Signs in with a token.

Desktop app

Windows/macOS shell that opens the same site in a window.

Signing in

Sessions expire after 20 minutes idle.
Cookies are renamed and locked to the site.
The admin page is not at /admin/.
The phone app uses a token, not a cookie.

THE SERVER (one Linode host)

nginx

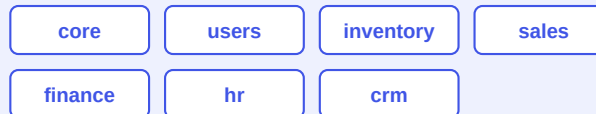
Terminates HTTPS, serves uploaded files, passes everything else back.

daphne — ASGI server

Runs Django. ASGI, not WSGI, because the app holds live connections open.

Django — the application

One app per part of the business:



Every app serves BOTH the web pages and the REST API — so the browser and the phone app always agree on the same rules and the same figures.

WHERE THINGS LIVE

PostgreSQL

Every product, sale, expense, employee and CRM record lives here.
It also keeps the history rows: who changed what, and when.

Live updates

When stock moves or a sale is made, the server pushes a message to every open screen — no refresh needed. Low stock raises an alert the same way.

Files on disk

Expense receipt photos, the company logo, employee photos, and the Android app that staff download from the login page.

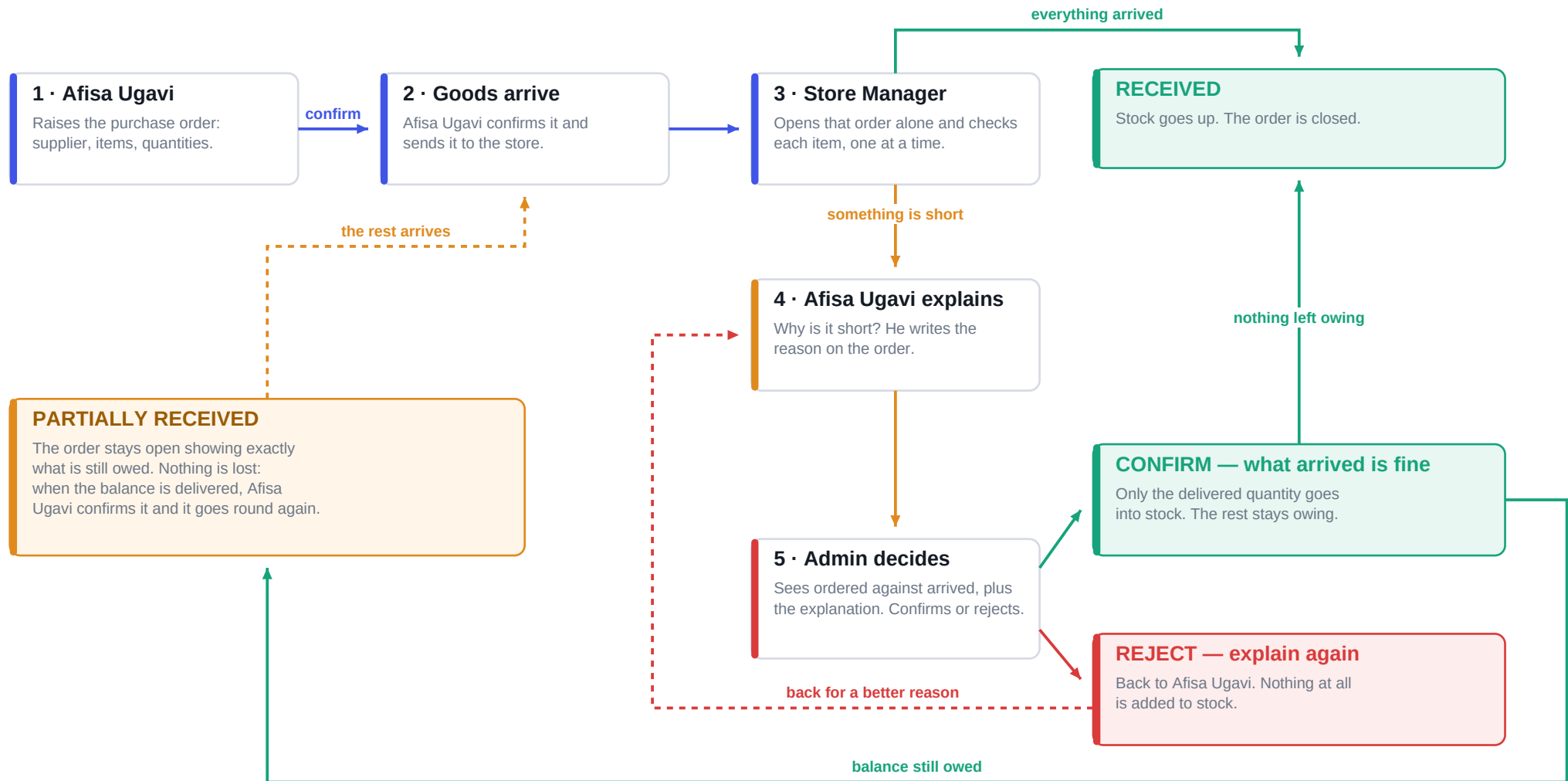
reads / writes

pushes

Why it matters: because the phone app and the web pages go through the same API, a rule fixed in one place is fixed everywhere. Nothing is written twice, so the two cannot disagree.

From purchase order to stock on the shelf

Three people sign off before goods count as stock. A short delivery cannot be waved through.

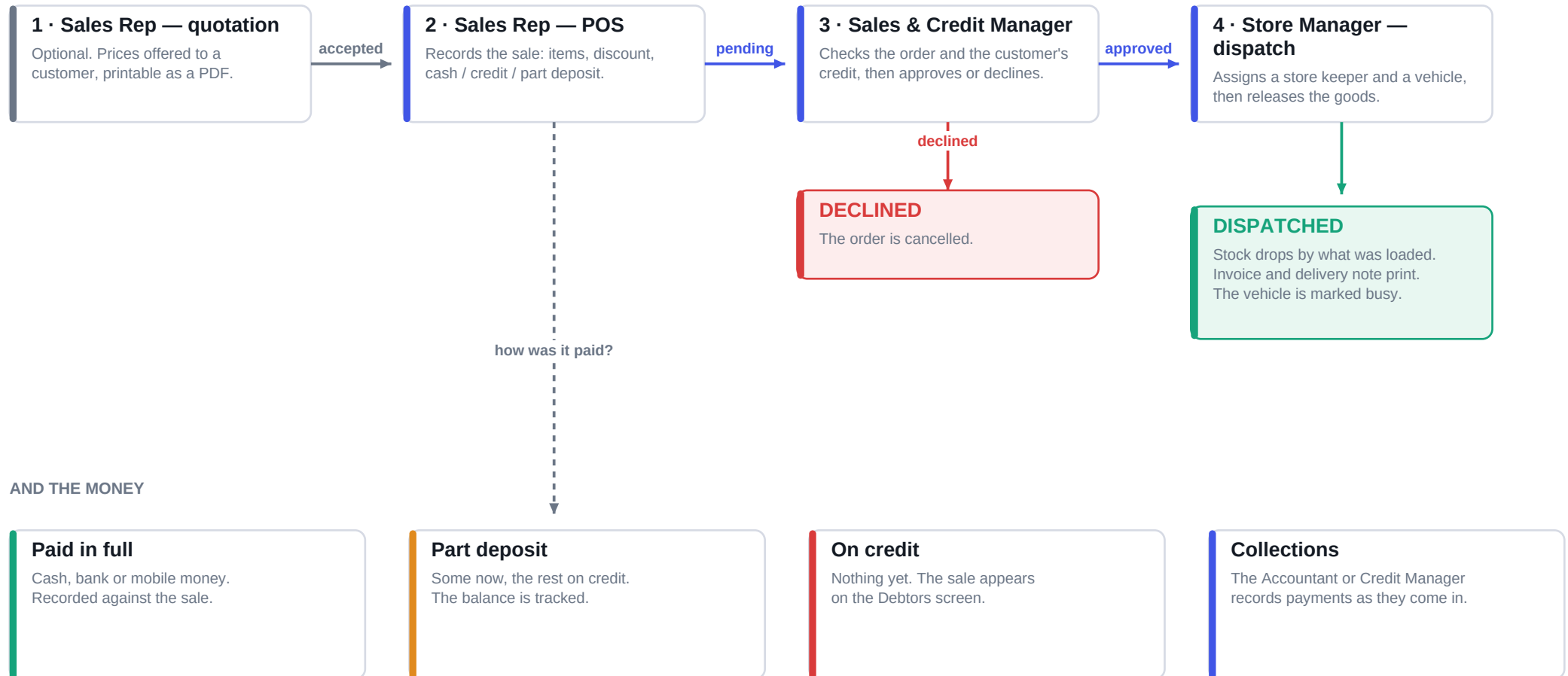


The rule that protects the stock figures: on a short delivery **NOTHING** is added to stock — not even the items that did arrive in full — until the Admin confirms it.

An order can be delivered in instalments. Every round keeps what was counted, who explained the shortfall and what the Admin decided, so the whole delivery history stays on the order.

From quotation to goods leaving the yard

A sale is raised by one person and released by another. Stock only drops when the lorry is loaded.



Two hands on every order: the person who raises the sale is never the person who releases it, and the goods are only deducted from stock at dispatch — so an order sitting unapproved never quietly drains the shelf.

Who does what

Signing in decides what you see. Each role opens on its own workspace and its own dashboard.

Afisa Ugavi Procurement • Raises purchase orders • Confirms deliveries arrived • Explains short deliveries • Suppliers and supplier payments • Trucks and transport costs	Store Manager The store floor • Checks deliveries item by item • Dispatches approved orders • Stock transfers between branches • Drivers and truck maintenance	Stock Controller Stock accuracy • Goods received notes • Stock balances and adjustments • Ageing and ABC reports
Store Keeper Physical goods • Verifies goods received • Physical stock counts • Loading and offloading	Sales Representative The shop counter • Quotations for customers • Records sales at the POS • Adds and looks after customers	Sales & Credit Manager Releasing and collecting • Approves or declines orders • Chases credit customers • Records payments received • Watches the debtors list
Accountant The money • Expenses, income and banking • Supplier payments and taxes • Customer receipts and debtors • The CRM customer register	HR Officer / Manager The people • Employee records and documents • Leave and attendance • Payroll and payslips • Performance and discipline	Admin Oversight • Confirms or rejects short deliveries • Creates users and sets roles • Sees every branch and every screen • Company settings

A person can hold more than one role — a manager who also keeps the books sees both workspaces. What each role may open is set once by the administrator, not screen by screen.